Practice Set — Unit II

Concepts in Metric Spaces

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Questions

1. **(Proof — Metric Axioms)** Let X be any nonempty set. Define

$$d(x,y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{if } x \neq y. \end{cases}$$

   Prove carefully that d satisfies all three metric axioms, and hence that (X, d) is a metric space.

2. **(Computation)** Let $x = (1, 3)$ and $y = (5, 0)$ in $\mathbb{R}^2$. Compute:

$$\text{(a) } d_1(x,y), \quad \text{(b) } d_2(x,y), \quad \text{(c) } d_\infty(x,y).$$

   Rank the three values and explain in one sentence why this ranking always holds for any two points in $\mathbb{R}^2$.

3. **(Visualization)** In $\mathbb{R}^2$, sketch and clearly label the unit ball $B_1(0)$ for each of the following metrics:

$$d_1, \qquad d_2, \qquad d_\infty.$$

   Then determine for each metric whether the point $P = (0.7, 0.6)$ belongs to $B_1(0)$. Justify your answer.

4. **(Computation)** Let $X = \{a, b, c\}$ with the discrete metric.

   (a) Find $B_1(a)$, $B_{1/2}(a)$, and $B_2(a)$.

(b) Find the sphere $S_1(a)$ and the closed ball $\overline{B}_1(a)$.

(c) Identify all interior points, all isolated points, and all limit points of the set $\{a, b\}$.

5. **(Proof)** Let (X, d) be a metric space, $x_0 \in X$, and $r > 0$. Prove that the closed ball

$$\overline{B}_r(x_0) = \{y \in X : d(x_0, y) \leq r\}$$

is a closed subset of X.

6. **(Proof)** Let (X, d) be a metric space and $x, y, z \in X$. Prove that

$$\big|d(x, z) - d(y, z)\big| \leq d(x, y).$$

Then explain in one sentence how this result shows that the function $f : X \to \mathbb{R}$ defined by $f(z) = d(x, z)$ is 1-Lipschitz.

7. **(Computation)** In $\mathbb{R}$ with the usual metric, find $\text{int}(A)$, $\overline{A}$, and ∂A for each set below. Briefly justify each answer.

(a) $A = (0, 1) \cup (1, 2)$

(b) $A = [0, 1) \cup \{3\}$

(c) $A = \mathbb{Q} \cap [0, 1]$

8. **(Proof)** Let (X, d) be a metric space.

(a) Prove that the union of any collection of open sets is open.

(b) Prove that the intersection of two open sets is open.

(c) Give an explicit example in $\mathbb{R}$ showing that the intersection of infinitely many open sets need not be open.

9. **(Visualization and Classification)** In $\mathbb{R}^2$ with the Euclidean metric, sketch the following sets and classify each as **open**, **closed**, **neither**, or **both**. Justify each classification.

(a) $A = \{(x, y) : x^2 + y^2 < 1\}$

(b) $B = \{(x, y) : x^2 + y^2 = 1\}$

(c) $C = \{(x, y) : x > 0,\ y > 0\}$

(d) $D = \{(x, y) : x \geq 0,\ y > 0\}$

10. **(Computation and Concept)** In $\mathbb{R}$ with the usual metric, find all limit points and all isolated points of each set.

(a) $A = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$

(b) $B = \mathbb{Z}$

(c) $C = \mathbb{Q} \cap (0, 1)$

For each set, state whether it is closed, and explain your reasoning in one line.

11. **(Proof)** Let X be a nonempty set with the discrete metric. Prove that:

(a) Every subset of X is open.

(b) Every subset of X is closed.

(c) X has no limit points.

12. **(Computation and Proof)**

(a) In $\mathbb{R}$ with the usual metric, compute $\operatorname{diam}(A)$ for $A = (0,1)$, $A = [0,1]$, and $A = \mathbb{Q} \cap [0,1]$.

(b) In the discrete metric space on any set X, compute $\operatorname{diam}(A)$ for any set A with $|A| \geq 2$.

(c) Prove that if $A \subseteq X$ is bounded and $B \subseteq A$, then B is also bounded.

13. **(Computation)** Let $Y = [0,1]$ be a subspace of $\mathbb{R}$ (with the usual metric).

(a) Is $[0, 1/2)$ relatively open in Y? Justify.

(b) Is $[0, 1/2]$ relatively closed in Y? Justify.

(c) Is $(1/3, 2/3]$ relatively open in Y? Justify.

(d) Is Y itself clopen in Y? Explain in one line.

14. **(Proof and Concept)** Let $X = \mathbb{R}$, and define on X:

$$d_1(x,y) = |x-y|, \qquad d_2(x,y) = \min\{1, |x-y|\}.$$

(a) Verify that d_2 is a metric on $\mathbb{R}$.

(b) Show that d_1 and d_2 are **topologically equivalent**, i.e., they generate the same collection of open sets.

(c) Show that d_1 and d_2 are **not Lipschitz equivalent** (so the stronger equivalence fails).

(d) Give one property that is preserved by $d_1 \leftrightarrow d_2$ and one property that is **not** preserved. Justify.

15. **(Proof — Extension)**

(a) On $\mathbb{R}^n$, prove the chain of inequalities

$$\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1 \leq n\,\|x\|_\infty,$$

where $\|x\|_p = \left(\sum_{k=1}^n |x_k|^p\right)^{1/p}$ for $p \in \{1,2\}$ and $\|x\|_\infty = \max_k |x_k|$.

(b) Conclude that d_1, d_2, d_∞ are **Lipschitz equivalent** on $\mathbb{R}^n$.

(c) Recall that ℓ^p denotes the space of sequences (x_n) with $\sum_n |x_n|^p < \infty$. Show that

$$\ell^1 \subsetneq \ell^2 \subsetneq \ell^\infty.$$

Solutions

1. **Solution.** We verify the three metric axioms.

   (i) Identity of Indiscernibles. If $x = y$, then $d(x, y) = 0$. Conversely, if $d(x, y) = 0$, then by definition $x = y$ (since when $x \neq y$, $d(x, y) = 1 \neq 0$).

   (ii) Symmetry. If $x = y$, $d(x, y) = d(y, x) = 0$. If $x \neq y$, then $y \neq x$, so $d(x, y) = 1 = d(y, x)$.

   (iii) Triangle Inequality. We need $d(x, z) \leq d(x, y) + d(y, z)$. Consider cases:

   - If $x = z$, then $d(x, z) = 0$, and the RHS is nonnegative, so we are done.
   - If $x \neq z$, then $d(x, z) = 1$. At least one of $x \neq y$ or $y \neq z$ must hold (otherwise $x = y = z$, contradicting $x \neq z$). So $d(x, y) + d(y, z) \geq 1$. Hence $d(x, z) = 1 \leq d(x, y) + d(y, z)$.

   All three axioms hold, so (X, d) is a metric space. □

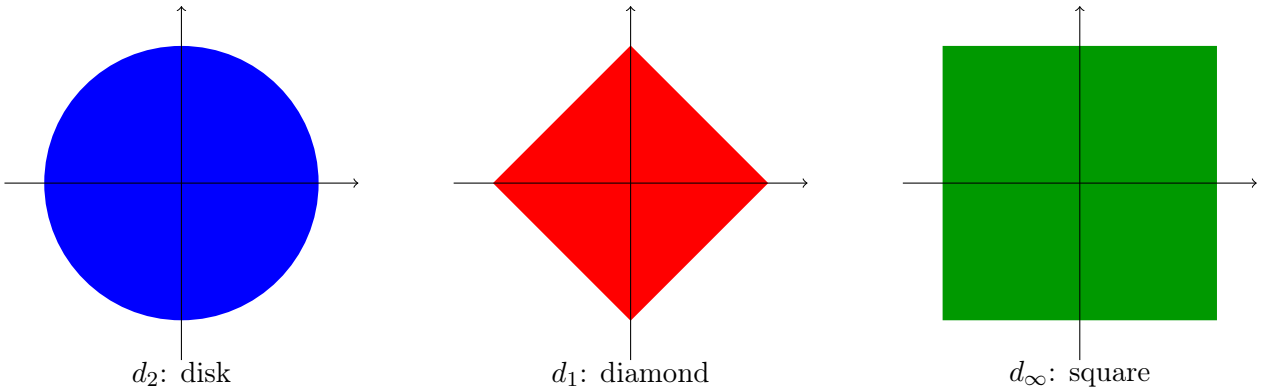
2. **Solution.** With $x = (1, 3)$ and $y = (5, 0)$:

$$\begin{aligned} d_1(x, y) &= |1 - 5| + |3 - 0| = 4 + 3 = 7, \\ d_2(x, y) &= \sqrt{(1 - 5)^2 + (3 - 0)^2} = \sqrt{16 + 9} = \sqrt{25} = 5, \\ d_\infty(x, y) &= \max\{|1 - 5|, |3 - 0|\} = \max\{4, 3\} = 4. \end{aligned}$$

   Ranking: $d_\infty \leq d_2 \leq d_1$, i.e., $4 \leq 5 \leq 7$.

   Why always? The inequalities $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ hold in $\mathbb{R}^n$ for every x (a special case of the general p-norm comparison). Geometrically: the supremum metric measures only the largest coordinate displacement (smallest); the Euclidean metric measures the true straight-line distance (middle); the taxicab metric measures the sum of coordinate displacements (largest). □

3. **Solution.** The three unit balls in $\mathbb{R}^2$ are:



d_2: disk d_1: diamond d_∞: square

   For $P = (0.7, 0.6)$:

   - d_2-norm: $\sqrt{0.49 + 0.36} = \sqrt{0.85} \approx 0.922 < 1$. So $P \in B_1(0)$ under d_2.
   - d_1-norm: $0.7 + 0.6 = 1.3 > 1$. So $P \notin B_1(0)$ under d_1.

- d_∞-norm: $\max\{0.7, 0.6\} = 0.7 < 1$. So $P \in B_1(0)$ under d_∞.

This illustrates that the *same* point can be inside one unit ball and outside another. □

4. **Solution.** Let $X = \{a, b, c\}$ with the discrete metric.

   (a) $B_1(a) = X = \{a, b, c\}$, since $d(a, b) = d(a, c) = 1 < r = 1$ (but *not* less than 1, so we need to check: $r = 1$ and $d < 1$? Points at distance exactly 1 are *not* in the open ball). Since $d(a, b) = 1$ is *not* < 1, we have $B_1(a) = \{a\}$. Wait — the discrete metric gives $d(a, b) = 1$ which is not less than 1, so only a itself is in $B_1(a)$. Corrected: $B_1(a) = \{a\}$. Similarly, $B_{1/2}(a) = \{a\}$, and $B_2(a) = X$ (since every distance is $\leq 1 < 2$).

   (b) Sphere: $S_1(a) = \{b, c\}$; closed ball: $\overline{B}_1(a) = X$.

   (c) For $A = \{a, b\}$:

   - Interior points: all of a and b (each is isolated in the discrete topology, and $\{a\}, \{b\} \subseteq A$ are open).
   - Isolated points: a and b.
   - Limit points: none (each point has $B_{1/2}(x) \cap (A \setminus \{x\}) = \emptyset$).

□

5. **Solution.** We prove $X \setminus \overline{B}_r(x_0)$ is open. Let $y \in X \setminus \overline{B}_r(x_0)$, so $d(x_0, y) > r$. Let $s = d(x_0, y) - r > 0$. Claim: $B_s(y) \subseteq X \setminus \overline{B}_r(x_0)$. Let $z \in B_s(y)$. By the reverse triangle inequality (Q6),

$$d(x_0, z) \geq d(x_0, y) - d(z, y) > d(x_0, y) - s = r.$$

So $z \notin \overline{B}_r(x_0)$, i.e., $z \in X \setminus \overline{B}_r(x_0)$. Hence $B_s(y)$ is contained in the complement. Therefore the complement is open, and $\overline{B}_r(x_0)$ is closed. □

6. **Solution.** By the triangle inequality,

$$d(x, z) \leq d(x, y) + d(y, z) \implies d(x, z) - d(y, z) \leq d(x, y).$$

Interchanging x and y:

$$d(y, z) \leq d(y, x) + d(x, z) = d(x, y) + d(x, z) \implies d(y, z) - d(x, z) \leq d(x, y).$$

Combining: $-d(x, y) \leq d(x, z) - d(y, z) \leq d(x, y)$, i.e., $|d(x, z) - d(y, z)| \leq d(x, y)$.

Consequence for Lipschitz. For fixed x, the function $f(z) = d(x, z)$ satisfies

$$|f(z) - f(w)| = |d(x, z) - d(x, w)| \leq d(z, w).$$

This is exactly the 1-Lipschitz condition. □

7. **Solution.**

   (a) $A = (0, 1) \cup (1, 2)$:

$$\text{int}(A) = (0, 1) \cup (1, 2), \quad \overline{A} = [0, 2], \quad \partial A = \{0, 1, 2\}.$$

   Reason: The set is already open, but 1 is a limit point not in the set, and 0, 2 are limit points.

(b) $A = [0,1) \cup \{3\}$:

$$\operatorname{int}(A) = (0,1), \quad \overline{A} = [0,1] \cup \{3\}, \quad \partial A = \{0,1,3\}.$$

Reason: 0 is not interior (no ball around 0 is inside A); 3 is isolated; 1 is a limit point.

(c) $A = \mathbb{Q} \cap [0,1]$:

$$\operatorname{int}(A) = \emptyset, \quad \overline{A} = [0,1], \quad \partial A = [0,1].$$

Reason: Any interval contains irrationals, so no point is interior; the irrationals of $[0,1]$ are limit points of $\mathbb{Q} \cap [0,1]$.

□

8. **Solution.**

(a) Arbitrary union. Let $\{U_\alpha\}_{\alpha \in I}$ be open sets, and $U = \bigcup_\alpha U_\alpha$. Take $x \in U$. Then $x \in U_\alpha$ for some α. Since U_α is open, there is $r > 0$ with $B_r(x) \subseteq U_\alpha \subseteq U$. Hence U is open.

(b) Finite intersection. Let U_1, U_2 be open, $U = U_1 \cap U_2$. Take $x \in U$. Since $x \in U_1$, there is $r_1 > 0$ with $B_{r_1}(x) \subseteq U_1$. Similarly, $B_{r_2}(x) \subseteq U_2$ for some $r_2 > 0$. Let $r = \min\{r_1, r_2\} > 0$. Then $B_r(x) \subseteq U_1 \cap U_2 = U$. So U is open.

(c) Infinite intersection counterexample. In $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty} (-1/n, 1/n) = \{0\},$$

which is not open (no ball around 0 is contained in $\{0\}$). □

9. **Solution.**



A: open disk B: circle C: open quadrant D: neither

(a) $A = \{(x,y) : x^2 + y^2 < 1\}$: **Open**. Every point has an open ball (a small disk) contained in A. Not closed (the boundary circle is missing).

(b) $B = \{(x,y) : x^2 + y^2 = 1\}$: **Closed**. Its complement $\{x^2+y^2 < 1\} \cup \{x^2+y^2 > 1\}$ is a union of open sets, hence open. Not open (no ball around a point of the circle is contained in the circle).

(c) $C = \{(x,y) : x > 0, y > 0\}$: **Open**. Same reasoning as (a). Not closed (the axes $x = 0$ and $y = 0$ are limit points not in C).

(d) $D = \{(x, y) : x \geq 0, y > 0\}$: **Neither**. Not open (points on the y-axis have no ball contained in D). Not closed (the line $y = 0$ with $x > 0$ consists of limit points not in D).

□

10. **Solution.**

(a) $A = \{1/n : n \in \mathbb{N}\}$: The only limit point is 0 (since $1/n \to 0$). Every element $1/n$ is isolated. So A is **not closed** (0 is missing).

(b) $B = \mathbb{Z}$: No limit points. Every integer is isolated. B is **closed** (contains all its limit points vacuously — there are none).

(c) $C = \mathbb{Q} \cap (0, 1)$: Limit points are $[0, 1]$ (irrationals in $[0, 1]$ are limits of rationals, and vice versa). No isolated points. C is **not closed** since $[0, 1] \not\subseteq C$.

□

11. **Solution.**

(a) Every subset is open. Let $A \subseteq X$ and $x \in A$. Then $B_{1/2}(x) = \{x\} \subseteq A$. Hence A is open.

(b) Every subset is closed. Let $A \subseteq X$. Its complement $X \setminus A$ is also a subset of X, hence open by (a). Therefore A is closed.

(c) No limit points. Let $x \in X$. Then $B_{1/2}(x) = \{x\}$, so $B_{1/2}(x) \cap (X \setminus \{x\}) = \emptyset$. Hence x is not a limit point of X. Since this holds for all x, X has no limit points. □

12. **Solution.**

(a) $\operatorname{diam}((0, 1)) = 1$, $\operatorname{diam}([0, 1]) = 1$, $\operatorname{diam}(\mathbb{Q} \cap [0, 1]) = 1$. (All three attain the same supremum.)

(b) In the discrete metric, $\operatorname{diam}(A) = 1$ if $|A| \geq 2$, and 0 if $|A| \leq 1$.

(c) If A is bounded, there exist $x_0 \in X$ and $r > 0$ with $A \subseteq B_r(x_0)$. Since $B \subseteq A \subseteq B_r(x_0)$, B is also contained in the ball $B_r(x_0)$, hence bounded. □

13. **Solution.**

(a) $[0, 1/2) = Y \cap (-1, 1/2)$, and $(-1, 1/2)$ is open in $\mathbb{R}$. Hence relatively open in Y.

(b) $[0, 1/2] = Y \cap (-1, 1/2]$, and $(-1, 1/2]$ is closed in $\mathbb{R}$. Hence relatively closed in Y.

(c) $(1/3, 2/3] = Y \cap (1/3, \infty)$ — wait, that gives $[1/3, ...]$; we need a different open set. Actually $(1/3, 2/3]$ is **not** relatively open in Y, because the point $2/3$ has no ball (in Y) contained in the set. To verify: any $r > 0$ yields $B_r^Y(2/3) = (2/3 - r, 2/3 + r) \cap [0, 1]$, which contains points just above $2/3$ — these are not in the set.

(d) Y is clopen in Y because Y itself is always both open and closed as a subset of any topological space (the empty set and the whole space are clopen by definition). □

14. **Solution.**

(a) *Non-negativity and identity:* $d_2(x,y) = \min\{1, |x-y|\} = 0 \iff |x-y| = 0 \iff x = y$. Symmetry is obvious. Triangle inequality: given x, y, z, if $d_2(x,y) = 1$, then $d_2(x,z) \le 1 \le d_2(x,y) + d_2(y,z)$. Otherwise $d_2(x,y) = |x-y|$. If $d_2(y,z) = 1$ or $d_2(x,z) = 1$, we are done. Otherwise all three are $|x-y|, |y-z|, |x-z|$ and the usual triangle inequality gives the result.

(b) *Topological equivalence.* It suffices to show that every d_1-ball contains a d_2-ball and vice versa. Given $x \in \mathbb{R}$ and $\epsilon > 0$:

- $B_\epsilon^{d_1}(x) = (x-\epsilon, x+\epsilon)$. Since $d_2 \le d_1$, we have $B_\epsilon^{d_1}(x) \subseteq B_\epsilon^{d_2}(x)$.
- Conversely, choose $\delta = \min\{\epsilon, 1\} > 0$. If $d_2(x,y) < \delta$, then since $\delta \le 1$, $d_2(x,y) = |x-y| < \delta \le \epsilon$. So $B_\delta^{d_2}(x) \subseteq B_\epsilon^{d_1}(x)$.

Hence the open sets coincide.

(c) *Not Lipschitz equivalent.* Suppose $d_1 \le C d_2$ for some $C > 0$. Take $x = 0, y = n$ large. Then $d_1(0,n) = n$ but $d_2(0,n) = 1$. So $n \le C \cdot 1 = C$ for all n, contradicting that n is arbitrary. Hence no such C exists.

(d) *Preserved / not preserved.* Preserved: open sets, closed sets, continuity, compactness (which is the same for both metrics). Not preserved: boundedness (under d_1, $\mathbb{R}$ is unbounded; under d_2, $\mathbb{R}$ is bounded by 1). $\square$

15. **Solution.**

(a) Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. Then:

- $\|x\|_\infty \le \|x\|_2$: Let k_0 be the index where $|x_{k_0}| = \|x\|_\infty$. Then $\|x\|_2^2 = \sum |x_k|^2 \ge |x_{k_0}|^2 = \|x\|_\infty^2$.
- $\|x\|_2 \le \|x\|_1$: $\|x\|_1^2 = \sum |x_k|^2 + 2\sum_{j<k} |x_j||x_k| \ge \sum |x_k|^2 = \|x\|_2^2$.
- $\|x\|_1 \le n\|x\|_\infty$: $\sum |x_k| \le \sum \|x\|_\infty = n\|x\|_\infty$.

(b) From (a), for $d_p(x,y) = \|x-y\|_p$, we get

$$d_\infty(x,y) \le d_2(x,y) \le d_1(x,y) \le n\, d_\infty(x,y).$$

So $\frac{1}{n} d_1 \le d_\infty \le d_1$, i.e., the metrics are Lipschitz equivalent with constants 1 and n.

(c) *Containment.* Let $x \in \ell^1$. Since $\sum |x_n| < \infty$, we have $x_n \to 0$, so $|x_n| \le 1$ for all $n \ge N$. Then for $p = 2$:

$$\sum_{n=1}^{\infty} |x_n|^2 = \sum_{n<N} |x_n|^2 + \sum_{n \ge N} |x_n|^2 \le \text{finite} + \sum_{n \ge N} |x_n| < \infty.$$

So $x \in \ell^2$. Same argument shows $\ell^2 \subseteq \ell^\infty$ (boundedness follows from convergence).

Strictness. The sequence $x_n = 1/n^{1/2}$ is in ℓ^2 (since $\sum 1/n^{3/2} < \infty$)? No, that's wrong. Let us use: $x_n = 1/n$. Then $\sum 1/n = \infty$ (so $x \notin \ell^1$), but $\sum 1/n^2 < \infty$ (so $x \in \ell^2$). This shows $\ell^1 \subsetneq \ell^2$. Similarly, $x_n = 1$ gives $x \in \ell^\infty$ but $x \notin \ell^2$, showing $\ell^2 \subsetneq \ell^\infty$. $\square$

— End of Practice Set —